Macroeconomics for Business – 2018 – Solutions

Reminder:

$$\text{Production function: } Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$$
$$\text{Income per capita: } y_t = \frac{Y_t}{N_t} = A_t k_t^{\alpha} \frac{L_t}{N_t}$$
$$\text{Growth decomposition: } g_t^y = g_t^A + \alpha g_t^K + (1-\alpha) g_t^L$$

1 Quiz

No negative points. Read the questions carefully. Only one answer is true.

1. Which of the following DOES NOT constitute **government expenditures** while accounting for GDP using the expenditure method?

   I. The government pays the salaries of military personnel

   II. The government collects a new sales tax on petroleum products that it will transfer to owners of electric vehicles

   III. The government purchases two trucks from Citroen that supplies them out of stock (produced last year).

   IV. The government purchases two newly produced ambulances from Renault

   ☐ I only ■ II only ☐ III only ☐ IV only ☐ I and II only ☐ I and III only ☐ I and IV only ☐ II and III only ☐ II and IV only ☐ III and IV only

2. Which of the following statements are true?

   I. Production value of black-market services is NOT registered in the French GDP.

   II. Exports (out of France) of German-owned firms located in France are included in the French GDP.

   III. Investment of French firms in foreign countries is included in the French GDP.

   IV. The wages of British expats employed in France counted towards French GDP

   ☐ I and II only ☐ I and III only ☐ I and IV only ☐ II and III only ☐ II and IV only ☐ III and IV only ☐ I, II and III only ■ I, II and IV only ☐ I, III and IV only ☐ II, III and IV only ☐ all of the above.

3. During the entire recession phase of the business cycle the economy can be in a positive output gap (pick the best answer)

   ☐ No, because a recession is defined by actual output being below potential output

   ☐ No, because the trough needs to be below potential output

   ☐ Yes, because the positive output gap is when output is below potential

   ■ Yes, if the recession is short-lived and current output does not fall below potential output before the next expansion

4. In Freedonia, housing starts fell, imports decreased and the consumer sentiment worsened considerably. You should definitely, in this phase of the cycle

   ☐ invest in stocks, as the return on equity will increase ■ move out of cash and stocks and buy bonds, as the relative returns to bonds versus these instruments should increase ☐ sell bonds, as a recession is highly likely now and there will be many bankruptcies ☐ get into cash to hold it safely over the coming recession

5. You observe, in Freedonia, that in a given year both the marginal product of labor and the marginal product of capital increased. What might have happened if you based your reasoning on a Cobb-Douglas production function (only one answer is correct!)?

   ☐ TFP fell.

   ☐ There was a strong increase in the number of hours worked.

   ■ TFP increased.

   ☐ There was an increase in capital (machines) available to workers in the economy.

6. Think about our model describing the overall economy. Suppose that firms invested, created and installed more capital that is ready to be used. What happens to wages (equal to the marginal product of labor) if the hours worked and labor utilization remain constant?

   ☐ They will fall because the share of capital in income will be higher.

   ■ Wages will increase because the marginal product of labor is increasing as more capital is available.

   ☐ Wages will stay constant.

   ☐ They will increase because TFP increases.

   For the following two questions. In some city, there are 100,000 inhabitants of working age. 50% of them either work or would like to work. Initially there is a 5% unemployment rate. Suppose that in this city a major employer restructures and 1,000 people are laid off that are currently unemployed (looking for work).

7. What is the **employment rate** in this city after this event?

   ☐ 46.5% ■ 93% ☐ 94% ☐ 95%

8. What is the **labor force participation** in this city after this event?

   ☐ 46.5% ☐ 47.5% ■ 50% ☐ 95%

9. Suppose you decide to hold 200 Euros less than usual in cash and deposit the entire 200 Euros in your checking account in a bank. What will be the most likely effect on the money supply M1 if banks do not keep excess reserves?

   ☐ No effect on M1 since cash is converted into checkable deposits.

   ☐ M1 will decrease since there is less cash in the hands of public (less cash in circulation).

   ■ M1 is likely to increase since banks will engage in multiple deposit creation.

   ☐ None of the above.

10. Suppose the central bank buys government bonds from commercial banks. Effectively, this move

□ increases the required reserve ratio

□ decreases the reserves of commercial banks

□ increases directly the overnight rate on the interbank market

■ expands the reserves of commercial banks

2 Growth accounting - The United Kingdom

Starting in the 90s, the financial sector in the United Kingdom became one of the most important one worldwide so that London was one of the main financial centers on the onset of the 2008 financial crisis. The development of this financial sector greatly contributed to economic growth in the UK. In this exercise, we want to understand how this development shaped British GDP and how these trends evolved with the financial crisis and afterward.

Time Period	Output growth	Capital growth	Labor growth
1999-2007	2.9%	1.9%	0.7%
2007-2009	-2.3%	1.2%	-0.9%
2009-2012	1.6%	1.0%	1.1%

Table 1: Average annual real growth (in percent) in United Kingdom.

(i) The labor share is estimated to be constant and close to 0.60 over the period 1999-2012. Calculate TFP growth rates for the three time periods 1999-2007, 2007-2009 and 2009-2012.

Answer:

$$TFP_{1999-2007} = 1.7\%, TFP_{2007-2009} = -2.2\%, TFP_{2009-2012} = 0.5$$

(ii) By comparing TFP growth between 1999-2007 and 2007-2009, explain how the development of the financial sector may have contributed to growth before and after the financial crisis?

Answer:

Basic: any comment saying that the development of finance was coincident with TFP growth for the period 1999-2007 and the financial crisis with a decline of TFP for the period 2007-2009.

More advanced: excessive dependence on one sector (finance).

(iii) After 2009, tighter regulations have limited the recovery of the financial sector, including in the UK. Based on your computation, what can you say about British long-term growth perspectives?

Answer:

Basic: reduced perspectives for finance implies reduced perspectives for the UK in terms of productivity gains and growth.

More advanced: a Dutch disease with finance?

3 Growth accounting – Tale of two neighbors

Poland and Ukraine are two neighboring countries that started their transition from planned to capitalist economies in 1990 at a roughly similar level.

Using data from Table **??** from Penn World Tables, answer the following questions. Assume that hours worked were fixed over the time period and that α, the capital share, is 0.4 for both economies for both 1990 and 2014.

	Poland		Ukraine	
	1990	2014	1990	2014
population (millions)	38.19526	38.61997	51.37001	45.0025
employed (millions)	15.63855	15.83779	25.29488	18.36319
human capital	2.732219	3.327214	2.905596	3.246984
share of labor compensation	0.6	0.6	0.6	0.6
Output (millions of 2011 $)	285787.8	937626.3	529948.1	461576.2
capital stock (current PPPs, millions of 2011 $)	864579.8	2073987	1788338	1586456

Table 2: National accounts – Poland and Ukraine

For the decomposition, start with the production function

$$Y = AK^{\alpha}(LH)^{1-\alpha}$$

where Y is output, A is TFP, K is the capital stock, L is the number of people employed while H is the human capital index. Then, using growth rates we can decompose it as

$$\ln\left(\frac{Y_{2014}}{Y_{1990}}\right) = \ln\left(\frac{A_{2014}}{A_{1990}}\right) + \alpha\ln\left(\frac{K_{2014}}{K_{1990}}\right) + (1-\alpha)\ln\left(\frac{L_{2014}}{L_{1990}}\right) + (1-\alpha)\ln\left(\frac{H_{2014}}{H_{1990}}\right)$$

1. Decompose growth of Real GDP for Poland and Ukraine over the time period 1990-2014. What is the contribution of capital accumulation, employment growth and the changes in education to GDP growth? What is the change in TFP for both countries over this time period? What is the most important contributor to growth for Poland and for Ukraine in terms of importance?

Answer: Note: the log-approximation here may not be the most accurate one for some items.

	Poland	Ukraine
$\ln\left(\frac{Y_{2014}}{Y_{1990}}\right)$	1.188	-0.138
$\alpha\ln\left(\frac{K_{2014}}{K_{1990}}\right)$	0.345	-0.048
$(1-\alpha)\ln\left(\frac{L_{2014}}{L_{1990}}\right)$	0.007	-0.192
$(1-\alpha)\ln\left(\frac{H_{2014}}{H_{1990}}\right)$	0.118	0.067
Log TFP growth	0.712	0.035

For Poland the most important factor driving growth was the increase in TFP, while for Ukraine it was the change (fall) in employment.

2. Calculate GDP per capita and GDP per worker for both countries for both years. What was the growth of GDP per capita in the period 1990-2014 in Poland and Ukraine? Which country has the highest labor productivity (GDP per worker) in 1990 and which one in 2014?

Answer:

	Poland		Ukraine	
	1990	2014	1990	2014
GDP per capita	7482	24278	10316	10256
GDP per worker	18274	**59201**	**20950**	25135
GDP per capita growth 2014-1990 (in pct)		224%		-0.5%
GDP per worker growth 2014-1990 (in pct)		223%		20%

3. What is the role of demography – the changes in the population – in the evolution of GDP per capita over the period for both countries (hint: decompose GDP per capita)?

$$\frac{Y}{N} = \frac{AK^{\alpha}(LH)^{1-\alpha}}{N}$$

Answer: we calculate

$$\ln \frac{\left(\frac{Y}{N}\right)_{2014}}{\left(\frac{Y}{N}\right)_{1990}} = \ln\left(\frac{Y_{2014}}{Y_{1990}}\right) - \ln\left(\frac{N_{2014}}{N_{1990}}\right)$$

	Poland	Ukraine
$\ln \frac{\left(\frac{Y}{N}\right)_{2014}}{\left(\frac{Y}{N}\right)_{1990}}$	1.177	-0.005
$\ln\left(\frac{Y_{2014}}{Y_{1990}}\right)$	1.188	-0.138
$\ln\left(\frac{N_{2014}}{N_{1990}}\right)$	0.011	-0.132

Note: careful on how the population is included in the calculation.

We see that there is little difference in output per capita growth and (aggregate) output growth for Poland given that the population barely changed in the period in comparison to the high change in GDP. For Ukraine, it matters: fall in aggregate output is matched by a large extent by the fall in the population and the resulting fall in employment (which was even higher than the fall in population as we calculated in the first point), and thus the GDP per capita barely moved.

4 Money (3 points)

Suppose Anne withdrew 200 euros in cash from her savings account (in Bank A). Bank A has sufficient cash in vault / excess reserves for this withdrawal without being forced to reduce its required reserves. Anne gave 50 euros to her daughter who then kept the cash in her piggy bank. Anne used the other 150 euros to buy food from Ben and Ben deposited the 150 euros cash into a checking account in Bank B.

1. Trace the changes in Bank A's and Bank B's balance sheet arising from the above transactions. (Write down the final changes only, not the intermediate steps). You may assume that the required reserves ratio is 10% on checkable deposits.

Bank A's balance sheet (changes only)

Assets	Liabilities
Reserves - 200	checkable deposits
Loans	other deposits - 200

Bank B's balance sheet (changes only)

Assets	Liabilities
Reserves +150	checkable deposits +150
Loans	other deposits

2. What is the net change of M1 money supply arising from the whole series of transactions? What about M2 money supply?

   Answer: M1 increases by 200 euros. M2 stays constant.

3. If Bank B did not want to keep excess reserves, what action is Bank B likely to take and what would be the consequence for money supply M1?

   Answer: Bank B can issue up to 150/(10%) - 150 = 1350 euros of new deposits.

4. Suppose now Ben bought a bond worth 100 euros from the central bank with a check drawn on his deposits at Bank B. Trace the changes in Bank's B balance sheet arising from this transaction.

   Bank B's balance sheet (changes only)

Assets	Liabilities
Reserves - 100	checkable deposits - 100
Loans	other deposits

5. What is the change in M1 money supply arising from this transaction?

   Answer: M1 decreases by 100 euros.

5 Business cycle

Based on the information below, comment on the following points:

- Where is currently the euro area in the cycle?
- In what types of assets (stocks or bonds) should you invest in? In which sectors?

Describe and define carefully each piece of information that you are using.

> Real GDP increased by 0.4%, quarter on quarter, in both the first and second quarters of the year, following the exceptionally strong dynamics observed in the previous five quarters. [...]
>
> In the second quarter employment rose further across euro area countries and sectors, increasing by 0.4% quarter on quarter. The average hours worked increased markedly in the second quarter, recovering from the decline due to the temporary impact of sick leave and strikes in the previous quarter. With the latest increase, employment stands 2.4% above the pre-crisis peak recorded in the first quarter of 2008. Since the trough recorded in the second quarter of 2013, cumulative employment growth in the euro area amounts to 9.2 million persons.
>
> [...] Private consumption rose by 0.2%, quarter on quarter, in the second quarter of 2018, following more dynamic growth in the previous quarter. This slowdown appears to reflect, on the one hand,

adverse temporary factors (such as the impact of strikes on transport-related spending in France) and, on the other hand, the unwinding of positive temporary factors in the first quarter (higher energy consumption due to the cold winter). On an annual basis, consumption rose by 1.3% in the second quarter of 2018, which represents a slowdown from the first quarter, when consumption rose by 1.6%. At the same time, annual growth of households' real disposable income increased from 1.7% in the first quarter to 1.9% in the second quarter. Thus, the slowdown in consumption growth was mirrored by a rise in the annual rate of change in savings, from 3.4% in the first quarter to 5.9% in the second quarter. The household saving ratio rose to 12.0% in the second quarter, slightly above the record low of 11.9% in the previous three quarters.

[...] As regards construction investment, households' increasing intentions to buy or renovate, as well as constructors' buoyant price and employment expectations, point to positive momentum in the construction sector over the short-term horizon. However, as financing conditions are expected to become slightly tighter, alternative long-term investment opportunities may gradually emerge.

Industrial production (excluding construction) rebounded and recorded a relatively strong increase in August, following the sharp declines of the previous two months. Still, on average, production in July and August stood 0.2% below the level seen in the second quarter of 2018, when it rose by 0.1% on a quarterly basis. This weakness partly relates to temporary bottlenecks in the production of cars triggered by the testing process following the introduction of the Worldwide Harmonised Light Vehicle Test Procedure (WLTP) on 1 September. More timely survey data signal ongoing robust growth at rates similar to, or somewhat lower than, those recorded in the first half of the year. The composite output PMI averaged 54.3 in the third quarter, compared with 54.7 in the second quarter, before declining in October to 52.7. Meanwhile, the European Commission's Economic Sentiment Indicator (ESI) eased to 112.5 in the third quarter, from 114.0 in the second quarter. Both the PMI and the ESI continue to stand above their respective long-term averages.

[...] The global composite output Purchasing Managers' Index (PMI) excluding the euro area declined further below its long-term average in September, reaching a two-year low. The decline was driven by both the manufacturing and the services components. In quarterly terms, the average PMI for the third quarter of 2018 declined from its level in the previous quarter. The composite output PMI decreased across most advanced economies, including the United States, Japan and the United Kingdom. In emerging market economies, it increased in India but weakened in China, Russia and Brazil, the latter country index remaining below the expansionary threshold.

Economic Bulletin – Issue 7 / 2018 – European Central Bank.

Note: the European Commission's ESI (Economic Sentiment Index) combines information from indicators on confidence in industry, services and financial services, consumer, construction and retail trade with the following weights: industry (40%), services (30%), consumers (20%), construction (5%) and retail trade (5%).

Answer keys:
Basic:

- You should carefully explain how each variable that they are using is reacting in the business cycle.
- The answer should be well structured and with clear arguments.
- In terms of content:
  - Coincident procyclical variables are increasing.

  - These variables (such as GDP) are increasing for a couple of quarters.
  - Advanced indicators are still on the positive side but decreasing.

- About investment decision, the only criterium is whether their conclusion regarding the cycle is consistent with their investment choice: So midcycle: still stocks, end cycle: cash becomes interesting (in general, people hold a more diversified portfolio).

The economy is in expansion, not necessarily at the very end of the expansion but we may experience a slowdown.

Advanced:

- The rest of the world economy, including key economies like the US, is much closer to recession given the PMI indicators. This will contribute negatively to the european business cycle.